

RotorLab — Build & Decision Log

A complete record of every technical decision made from geometry through solver launch: what was chosen, why, and every failure mode hit along the way. Companion to the CFD Guide (physics/theory) and the Rebuild Runbook (quick rebuild checklist) — this document is the full narrative behind both.

0. What This Project Is

A standard Rushton turbine stirred tank ($T = 0.300$ m), built entirely in 3D-CAD with no imported or exported CAD files, comparing steady MRF against transient sliding mesh, both benchmarked against the published Power Number $Po \approx 5.0$ for this geometry. Everything below is in the order it actually happened, including the wrong turns.

1. Geometry

1.1 Dimensions (standard Rushton, $T = 0.300$ m)

Part	Value
Tank diameter T / Sketch 1 radius	0.300 m / 0.150 m
Liquid height H	0.300 m
Impeller diameter D	0.100 m
Off-bottom clearance C (disc Z)	0.100 m
Disc diameter / thickness	0.075 m / 0.002 m
Blade length / height / thickness	0.025 / 0.020 / 0.002 m
Blades / Baffles	6 at 60° / 4 at 90°
Baffle width / Shaft diameter	0.030 m / 0.015 m
RotZone radius / height / true center Z	0.081 m / 0.060 m / 0.103 m

1.2 Build-order rules that mattered

Body Interaction = None on every new body, set immediately on creation. Left at its default (Merge, All), a new extrude silently fuses into whatever body already occupies that space — specifically TankFluid, since every solid body was built inside it.

Blade-disc overlap. Placing a blade exactly tangent to the disc's outer radius (single-point contact, zero volumetric overlap) makes Unite fail with “not all tool bodies are united into target body.” Fix: extend each blade about 1 mm past the disc's true radius so there is genuine shared volume, not just contact.

Baffles were never meant to Unite. Four baffles spaced 90° apart never touch each other, so a Unite between them is topologically impossible — that failure is correct behavior, not a bug. They're used as separate Subtract tool bodies instead.

RotZone's “Center” field. This one cost the most time and gets its own section (5) — short version: the field labeled “Center” on the Cylinder primitive is actually the center of the base circle before extrusion, not the volumetric center of the resulting solid.

1.3 Splitting one tank into two fluid volumes

There's no dedicated Split tool in this version, so the rotating/stationary split is built from ordinary Booleans: `Subtract(Target = TankFluid, Tools = [Impeller, 4 Baffles])` removes all solids at once, giving one combined fluid body (HollowedTank). Because Intersect and Subtract both consume every body they touch, and RotZone is needed as an input to two separate downstream operations, **both** HollowedTank and RotZone have to be duplicated (`Ctrl+C/Ctrl+V`) before either operation runs — not just HollowedTank. Then `Intersect(HollowedTank, RotZone)` keeps only the overlap (RotatingFluid), and `Subtract(HollowedTank copy, RotZone copy)` removes that same overlap from the other copy (StationaryFluid). This works because `Intersect(A,B)` and `Subtract(A,B)` together exactly partition A — no gap, no overlap — but only if both operations act on genuinely identical, independent copies of both A and B. The first version of the rebuild runbook missed duplicating RotZone and only caught it on review.

2. Parts, Regions, Interface

2.1 New Geometry Part

Right-click 3D-CAD Model 1 -> New Geometry Part. In this project it did far more than create Parts: it auto-detected 4 pairs of coincident faces between RotatingFluid and StationaryFluid, built a Part Contact from them, assigned both Parts to new Regions, assigned their surfaces to boundaries, and built a correctly-typed Interface — all in one action.

2.2 Why interface type matters

A Direct Region Interface (either Region-Mode or Contact-Mode, built by selecting whole Regions) comes in as a Heat Exchanger type — thermal-only coupling that blocks flow between the regions. Wrong for a rotating/stationary fluid split. The correct type — Type: Internal Interface, Topology: In-place, Connectivity: Imprinted — comes from selecting individual **boundaries** (Contact-Mode Boundary Interface), or, as happened here, from the automatic Part-Contact detection inside New Geometry Part.

2.3 Verification logic used

Each Region's own **Parts** property was checked directly (Assign Parts to Regions doesn't always populate it). Each boundary's Part Surfaces field showing <Region>.Default was treated as proof by elimination: Default means “everything not claimed by a more specific split,” and since the interface is the only other split taken out of each region, Default is logically forced to be the correct wall surface — regardless of how it looks from one camera angle in a render.

3. Meshing

3.1 Two separate Automated Mesh operations

The Trimmed Cell Mesher in this version cannot process multiple Parts inside a single Automated Mesh operation (“Trimmer cannot currently accommodate multiple parts in one AutoMesh Operation”). Fix: one operation per Part (RotatingFluid, StationaryFluid) rather than one operation with both as Input Parts. This also simplified the impeller's fine refinement — since that operation is now scoped to one Part, the finer Target/Minimum surface size can be set directly in its own Default Controls instead of needing a separate Custom Control override.

Setting	Both operations	RotatingFluid only
Mesh models	Surface Remesher + Trimmed Cell Mesher + Prism Layer Mesher	same
Base size	0.010 m	same
Target / Min surface size	defaults	1.5 mm / 0.5 mm
Prism layers / stretching	12 / 1.2	same
Prism total thickness	0.0008 m — Absolute, not Relative	same

3.2 What “100% face validity” doesn't tell you

A mesh can report perfect face validity everywhere and still represent the wrong physical domain — validity measures whether cells are well-formed, not whether the rotating/stationary split is actually where it's supposed to be. That distinction is exactly what Section 5 turned on. The reliable cross-check used throughout: total boundary faces (ImpellerWall + TankWalls) matching the Overall summary's boundary face count exactly, confirming the Imprinted interface's faces are correctly absorbed into interior connectivity rather than double-counted or missing; and direct body **Extents** (Min/Max), which is the check that actually caught the RotZone defect that validity checks could not.

4. File Loss & Persistence

Mid-project, the working .sim file was saved and closed normally, then would not reopen: STAR-CCM+'s own loader reported “corrupted, Invalid begin of file,” and the file was tens of KB instead of the expected tens-to-hundreds of MB — a write truncated by a crash, full disk, or (most likely) a non-persistent virtual desktop resetting local storage between sessions. No recovery was possible: Windows Previous Versions was empty and the VDI had no web/OneDrive access. Full geometry and mesh rebuild followed. Lesson kept from then on: **Auto Save only protects the solver-iteration phase** — it does nothing during 3D-CAD or meshing, since nothing is iterating yet. The actual protection for that phase is a confirmed persistent drive plus manual, versioned Save-As at fixed checkpoints.

5. Case Study — the RotZone Rebuild

After the full rebuild, mesh diagnostics showed RotatingFluid/ImpellerWall spanning $Z = [0.103, 0.163]$ — but the disc sits at $Z=0.100$ and the blades span roughly $Z=0.090-0.110$, so most of the true impeller fell outside the meshed rotating zone entirely. Two wrong theories came before the right one:

Wrong theory 1 — base vs. center confusion. Suspected the Cylinder primitive's Z value had been entered as a base rather than a center. Checked Cylinder 1's actual Properties directly: Center = [0,0,0.103], Radius 0.081, Height 0.06 — correctly centered, by its own label. Theory dropped, incorrectly.

Wrong theory 2 — stale duplicate chain. Suspected the Intersect/Subtract chain had consumed a RotZone snapshot taken before some earlier correction, and that "Update 3D-CAD" wasn't forcing it to regenerate. Deleted the entire downstream chain (DuplicateBody, IntersectBodies, SubtractBodies) and rebuilt it fresh from the current Cylinder 1 output. Result: bit-for-bit **identical** Volume and Extents to before the rebuild. That identical result is what disproved the theory — if the chain had been stale, a fresh rebuild would have produced different numbers. It didn't, which meant the chain was never broken; it was faithfully reproducing exactly what Cylinder 1 actually was.

Root cause. The field labeled "Center" on this Cylinder primitive is actually the center of the 2D base profile (the circle, before extrusion) — Height then extrudes upward from there, not symmetrically around it. Entering Center Z=0.103 produced a solid spanning 0.103 to 0.163, not centered on 0.103 at all. A legitimate, if misleadingly named, convention in sketch-then-extrude primitive tools.

Fix. Edited Cylinder 1 and Cylinder 2 directly: Center Z changed from 0.103 to 0.073 (= 0.103 – half the 0.060 m height). Radius and Height untouched. Verified via a fresh body-level Volume/Extents measurement on RotatingFluid: Min [-0.081,-0.081,0.073], Max [0.081,0.081,0.133] — exactly the intended span, true center landing back on 0.103.

The lesson worth keeping: a Volume reading alone (0.0012160 m³) looked plausible both before and after the fix, and was nearly identical in both the broken and corrected states — volume can't distinguish a correctly-positioned body from a wrong one that happens to be a similar size. Extents (Min/Max) was the only check that actually discriminated between right and wrong here. And a field's on-screen label ("Center") isn't proof of its actual behavior — verify against measured geometry, not against what a UI calls something.

6. Physics Continuum

Model	Why
Steady	MRF is a steady approximation of rotation, not a moving-mesh transient
Segregated Flow	standard, efficient choice for incompressible single-phase liquid
Constant Density	liquid, no compressibility effects expected
Turbulent -> RANS -> Realizable K-Epsilon Two-Layer	robust, well-validated standard for rotating-machinery / mixing-tank CFD
Two-Layer All Y+ Wall Treatment	matches the prism layer sizing target ($y^+ \approx 20-30$), works across the y^+ range
Wall Distance, Gradients	auto-added dependencies of the models above, expected

Material overridden to **Water: density 998 kg/m³, dynamic viscosity 0.001 Pa·s** — matching the guide's spec exactly, since Reynolds number and Power Number are both directly sensitive to these two numbers.

7. Motion & Reference Frame

Rotation motion (Tools -> Motions -> New -> Rotation): Rate 300 rpm (31.416 rad/s), Axis Direction [0,0,1], Axis Origin [0,0,0] — the tank's vertical centerline, which every body in this project was built symmetric about.

The Motion vs. Reference Frame distinction. On RotatingFluid's Physics Values -> Motion Specification, the correct combination is **Motion: Stationary + Reference Frame: [the Rotation reference frame]** — not Motion set to Rotation. This is exactly what MRF means physically: the mesh itself never moves; only the reference frame the flow equations are solved in rotates, adding the Coriolis and centrifugal source terms mathematically. StationaryFluid was left untouched at Lab Reference Frame.

ImpellerWall's own Physics Conditions confirmed **Reference Frame Specification: Region Reference Frame** — meaning the wall automatically follows whatever reference frame its parent region has, making it a standard no-slip wall relative to the rotating frame, which correctly represents a solid impeller physically spinning in the lab frame. No override needed.

Decision: top boundary left as a wall, not split out as a Symmetry Plane. Power Number is driven almost entirely by near-impeller torque (blade drag, trailing vortices); the top boundary mainly affects far-field free-surface behavior, which has minor influence on the benchmark metric in a baffled tank where bulk swirl is already suppressed. Flagged as a possible sensitivity study for later, not required now.

8. Reports, Monitors, Stopping Criteria

Moment 1 report field	Value / reasoning
Parts	RotatingFluid: ImpellerWall
Axis / Axis Origin	[0,0,1] / [0,0,0] — matches the Motion exactly
Force Option	Pressure + Shear (torque needs both contributions)
Reference Pressure	0.0 Pa — doesn't actually matter here: a uniform pressure offset contributes zero net moment over a closed surface, and ImpellerWall fully encloses the solid impeller
Units / Coordinate System	N-m / Laboratory

A Monitor + Plot was created from Moment 1 to watch torque vs. iteration directly, rather than trusting residuals alone. Power Number is computed by hand from the converged torque afterward rather than as a live in-solver expression — a one-line conversion doesn't need extra setup: $Po = \text{Torque} \times 2\pi N / (\rho N^3 D^5)$, which for $N=5$ rev/s, $D=0.1$ m, $\rho=998$ kg/m³ reduces to $Po \approx \text{Torque} \times 25.18$.

Stopping Criteria: **Maximum Steps = 1500** (Enabled, Logical Rule Or) as an unconditional safety net. The Moment monitor criterion was initially set to Criterion Option **Minimum** by default — wrong semantics: that's a threshold-crossing check (stop once a value drops below some limit), not a stability check, and risked stopping the run prematurely if torque dipped near an arbitrary default limit early on. Changed to **Asymptotic**. Its own defaults were also too loose to trust: $|\text{Max}-\text{Min}| = 0.1$ with **Normalized** checked (a ~10% relative band around the monitored value — loose against an expected torque near 0.2 N·m) and only 10 samples in the checking window. Tightened to $|\text{Max}-\text{Min}| = 0.02$ ($\approx 2\%$) over **50 samples**, so the run won't be declared converged on a short flat stretch or ordinary iteration noise.

9. Visualizing the Flow

Safe to do while the solver is actively running — creating scenes and derived parts doesn't require pausing or stopping a run.

Cutting plane. Derived Parts -> New Part -> Section -> Plane, Input Parts = both regions. A plane is defined by a normal direction plus one coordinate along that normal — Normal [1,0,0] with Origin.x = 0 gives a vertical slice through the tank's centerline; the Origin.y and Origin.z values don't actually change which plane results, since the tank is axisymmetric about Z and the plane extends infinitely regardless of those two numbers. This gives the classic side-view of the impeller's radial jet splitting into upper and lower recirculation loops.

In the plane-creation dialog's own Display section, choosing **New Scalar Displayer** (instead of the default New Surface Displayer) creates the colored contour directly, skipping a separate New Scene step.

Scalar Field: Velocity -> Lab Reference Frame -> Magnitude — not ReferenceFrame for Rotation. Because RotatingFluid's equations are solved relative to the spinning frame, STAR-CCM+ can report velocity in either frame. "ReferenceFrame for Rotation" shows the flow as seen by an observer spinning with the impeller (the blades would look stationary) — a specialized view for blade-relative features, not a general one. "Lab Reference Frame" gives the absolute velocity actually observed from outside the tank, and is also the only choice that blends smoothly across the RotatingFluid/StationaryFluid interface — StationaryFluid's velocity is naturally reported in the lab frame, so plotting RotatingFluid in the rotating frame instead would create a visually discontinuous jump right at the interface line.

10. Where This Stands

Steady MRF run launched with the setup above. Next: extract converged torque and Power Number once it plateaus, then build and run the transient sliding-mesh case on the same geometry, compare both methods against each other and against the $Po \approx 5.0$ published benchmark, then write up the comparison and push to GitHub.